

Practice Quiz: Learning

Part A: Multiple Choice

1. In Active Inference, parameter learning refers to: A) Adding new variables to the generative model B) Updating concentration parameters (e.g., Dirichlet priors) based on experience C) Discarding the generative model and starting from scratch D) Learning the laws of physics through formal education
2. Structure learning (Bayesian Model Reduction) involves: A) Refining predictions within an existing model structure B) Analytically comparing nested models and selecting the one with best evidence C) Building physical structures in the environment D) Learning only when explicitly told to by a teacher
3. Niche construction in the context of Active Inference refers to: A) Building a nest or shelter for physical protection B) The agent acting on its environment to make it more predictable, reducing future surprise C) The environment changing the agent's preferences D) A purely biological process irrelevant to philosophy
4. The concept of *Bildung* in German philosophy refers to: A) Construction of buildings and physical infrastructure B) Self-formation through education — becoming a different kind of person through learning C) Memorization of cultural facts D) Biological maturation of the nervous system
5. The "complexity" term in Bayesian model evidence: A) Favors the most complex model available B) Penalizes unnecessary complexity, implementing a formal version of Occam's Razor C) Measures the computational cost of running the model D) Is irrelevant to learning
6. If learning transforms the generative model, and the agent *is* its generative model, then: A) Learning changes the agent's identity, not just its knowledge B) Learning is impossible because the agent cannot change itself C) The agent must be separate from its model D) Personal identity is determined at birth and cannot change

7. Kuhn's concept of a "paradigm shift" is most closely analogous to: A) Parameter learning — refining existing beliefs B) Structure learning — changing the model's architecture to better explain evidence C) Forgetting — losing previously acquired knowledge D) Perception — seeing something for the first time

Part B: Short Answer

1. Explain how Bayesian Model Reduction (BMR) implements a formal version of Occam's Razor. Why might simpler models be preferred even when more complex models fit the data better?
2. A child who has never seen snow experiences it for the first time. Describe this experience using the Active Inference framework: is this parameter learning, structure learning, or both? What happens to the child's generative model?
3. Construct an argument for or against the claim: "If I am my generative model and learning changes my model, then the person I was before a transformative experience no longer exists."